WasteWise

Predictive Waste Intelligence for Smart Cities

An AI-powered municipal solid waste management system that combines computer vision, forecasting, and route optimization to cut unnecessary trips, prevent overflows, and save fuel.

Open-source prototype | Synthetic data demonstration

GitHub: https://github.com/KananSangal/WasteWise-Prototype

Agenda

01 **The Challenge**
Why fixed-schedule waste collection fails Indian cities

02 **What WasteWise Does**
Overview of the predictive intelligence layer

03 **Core Capabilities**
Dashboard, forecasting, routing, and computer vision

04 **How It Works**
Architecture and technology stack

05 **Simulated Impact**
What the models show on synthetic Jaipur data

06 **Status & Next Steps**
What is built, what is planned, and honest limitations

The Challenge in Indian Cities

62 Mt

Municipal solid waste generated annually in India

~70%

Of collected waste still ends up untreated or dumped

Fixed

Routes ignore actual bin fill levels every day

What this means on the ground

Wasted trips

Trucks visit near-empty bins while others overflow

Overflow risk

Especially near markets and during festivals

Higher emissions

Extra fuel burned and more CO_2 released

WasteWise: A Predictive Intelligence Layer

WasteWise sits on top of existing bins and trucks. It does not require replacing every bin overnight. Instead it adds three smart capabilities:

See

Estimate how full each bin is (computer vision on images or future sensors)

Predict

Forecast fill levels hours and days ahead so crews know where problems will appear

Optimise

Send trucks only to bins that actually need emptying, with capacity-aware routes

Computer Vision for Bin Fill Estimation



Conceptual illustration – smart multi-bin station with camera

What the current model does

A lightweight CNN looks at a photo of a bin and estimates:

- Fill class (Empty / Low / Medium / Full)
- Exact fill percentage (0–100%)

Why it matters

Knowing fill level without sending a person or truck is the foundation for demand-driven collection. The current prototype uses synthetic images; real camera or ultrasonic sensors can plug into the same pipeline.

Intelligent Collection Planning



Example neighbourhood layout with numbered collection points (conceptual)

Forecasting Engine

Two complementary models work together:

Prophet (7-day zone view)

Captures weekly patterns and Indian holidays. Helps planners see which zones will be busy next week.

XGBoost (6-hour bin view)

Uses recent fill history, time of day and bin type to predict the next few hours. Flags bins that will cross the collection threshold soon.

Smart Route Optimisation & Control View



Conceptual control-tower dashboard (illustrative)

How routing works

- Only bins ≥ 70% full are considered
- Greedy nearest-neighbour heuristic
- 3 trucks, each with 2,000 L capacity
- Trucks return to depot when full
- Haversine distance on real GPS points
- Side-by-side comparison vs fixed routes

From Data to Action on the Street



Conceptual view of connected smart bins and collection vehicle with real-time data overlay

System Architecture (as implemented)

Frontend	ML Models	Optimisation	Data
Streamlit multi-page app Dashboard • Route Map • Forecasting • CV Demo	Prophet (zone 7-day) XGBoost (bin 6-hour) TensorFlow CNN (fill class + %)	Nearest-neighbour VRP Capacity constraints 70% fill threshold	100 synthetic bins (5 zones) 30 days × 24 h fill history Jaipur GPS coordinates

Technology Stack

Language & Runtime

Python 3.10+

Web Framework

Streamlit

Deep Learning

TensorFlow / Keras

Forecasting

Prophet, XGBoost

Mapping

Folium (Leaflet)

Charts

Plotly

Data

Pandas, NumPy

Deployment

Local / any Python host

Simulated Impact (Synthetic Jaipur Data)

These figures come from the project's own simulation on 100 bins and a 3-truck fleet. They illustrate potential, not guaranteed real-world results.

34%

Fewer collection trips

vs fixed daily routes

~30%

Shorter total distance

route efficiency gain

47 L

Fuel saved per day

example fleet scale

12

Overflows prevented / week

proactive collection

Source: Project README simulated impact table. Real cities typically report 15–40% distance reductions with fill-level driven routing (literature range).

What Is Built vs Planned

Implemented (working code)

- ✓ Synthetic data generator (100 bins, 30 days)
- ✓ Prophet + XGBoost training scripts
- ✓ CNN bin-fill classifier & regressor
- ✓ Nearest-neighbour route optimiser
- ✓ Full Streamlit multi-page dashboard
- ✓ Interactive Folium route maps
- ✓ KPI cards, trends, heatmaps

Not yet / Future work

- ○ Real IoT ultrasonic / camera feeds
- ○ Live production deployment
- ○ Multi-class waste-type sorting (plastic, food…)
- ○ Mobile driver app
- ○ Integration with existing municipal systems
- ○ User authentication & role management
- ○ Large-scale real-city pilot

Honest Limitations & Assumptions

Synthetic data only

All fill levels and images are generated. Real sensor noise, camera conditions and human behaviour are not yet modelled.

Simplified routing

Nearest-neighbour heuristic is fast but not optimal. Real fleets need traffic, time windows and labour rules.

No waste-type sorting

Current CNN estimates fill level, not residual / plastic / food / glass. Multi-class sorting is a natural next step.

Single-city focus

Bin locations and holiday calendars are Jaipur-specific. Transfer to other cities requires new data.

No live operations

There is no production deployment or integration with municipal control rooms at present.

Impact numbers are simulated

Reported savings illustrate model behaviour on synthetic data; field trials are required for validation.

Key Sources & References

1. WasteWise repository: https://github.com/KananSangal/WasteWise-Prototype

2. India MSW generation ≈ 62 million tonnes/year – CPCB / TERI / MoEFCC consistent estimates

3. Route-optimisation literature: 15–40% distance reduction with fill-level driven collection (multiple peer-reviewed studies)

4. IoT smart-bin meta-analysis: average ~21% collection distance reduction (MDPI, 2025)

5. Project simulated impact numbers taken directly from the repository README

6. Prophet (Facebook/Meta) and XGBoost documentation for forecasting methodology



WasteWise

Turning fixed-schedule collection into demand-driven intelligence

Open source • Extensible • Built for real municipal problems

Github Prototype Link

Questions welcome